# Movie Social Network Attributes as Predictors of Box Office Success

Jordan Conrad-Burton

*Department of Software and Information Systems Engineering, Ben-Gurion University of the Negev*

April 2024

**Abstract**

The topic of predicting the success of a movie based on certain features has been a subject of interest for decades. These features range from social media and sentiment analysis to individual behavior analysis to even cinema arrangements. This study looks at a new subset of features to consider: the attributes of the social network of the characters in the movie. This analysis indicates that there is indeed an effect of a film's social network attributes on the level of success the movie experiences both financially and critically, introducing a possible new aspect of consideration in the widely researched domain of movie success prediction.

## 1 Introduction

The application of data science tools and machine learning has advanced many fields, enabling us to understand and analyze trends that previously went unseen. One domain of application, in large part due to the availability of massive datasets, is that of movie box office prediction. Forecasting the success (or failure) of a movie at the box office is a problem that has interested both researchers as well as those associated with the film industry itself. After all, being able to accurately predict the success of your movie before it even hits theaters would be a useful and profitable tool.

This study is an analysis of the effects of the social network of the movie itself, i.e. of the characters in the film, on the eventual success of the movie both in terms of box office profit and the rating of the film. Films are classified into three different levels of success based on their rating and their ratio of gross worldwide revenue to budget: failure, partially successful, and successful. Using this analysis, we can point out certain characteristics in how the interactions between characters are built in the film can affect the overall success of the movie.

The remainder of this paper is organized as follows: Section 2 is a brief overview of related work to this study; in Section 3, methods, datasets, and analysis are presented; Section 4 contains the results of said analysis, and Section 5 conclusions from this study are given along with suggestions for future work.

## 2 Related Works

Most models created to predict the box office success of a movie fall into one of two categories based on their primary focus: behavioral models that focus on an individual's decisions in choosing a movie or quantitative models that focus on factors that influence box office profits of newly released movies. Another classification of these models is the timing in which they predict the success or failure of a movie, either before the initial release or afterwards [1].

Previous studies have focused on different aspects of the movie industry as a way to predict the financial success of a film - from the marketing and distribution pattern and cinema arrangement [2] to sentiment analysis using social media websites and movie reviews [3]. There are even studies that focus on predicting box office markets in different countries [4].

However, a previously unexplored domain of predictive features seems to be an analysis of the movie itself, specifically its social network characteristics. This study differentiates itself from previous works in that it provides an initial analysis of the impact of the construction of a movie's social network on that movie's eventual success both financially and critically in order to ascertain whether or not this is a viable angle from which to analyze the topic of movie success.

# 3 Methods

## 3.1 Social Network Features

The dataset of social networks of different movies compiled in the study "Using data science to understand the film industry's gender gap" [5] was used in order to extract the social network attributes. The social networks in this dataset were constructed according to the appearance of characters in the movie. Given a movie, a graph G = ⟨V,E⟩was constructed where the characters in the movie are the nodes. The edges were defined as the interaction between the two characters with a weight that represents the number of interactions between them. An edge between two characters was created if they appeared in the movie in a time interval of less than 60 seconds. For each appearance, the weight of the edge between them was increased by one. All edges with a weight lower than 3 were filtered out to account for possible false positive scenarios.

The data from these social networks was compiled and certain attributes were extracted for use in this study. The specific social network attributes analyzed and their definitions are as follows:

- *Number of movie characters*: the number of nodes in the movie network.
- *Network closeness*: the number of edges in the movie network. The more edges in a network, the "closer" the characters are to each other since they interact more with each other.
- *Number of connected components*: number of separate sub-networks within the entire social network.
- *Number of main characters*: a "main" character of a social network is defined as a node that has a degree centrality greater than 0.6.
- *Node with maximum degree*: this is equivalent to the character in the movie who interacts with the largest number of different characters.
- *Edge with maximum weight*: this is equivalent to the two characters who interact the most with each other in the movie.
- *Network strength*: the total sum of the weights of the edges in the network, meaning the total amount of interactions between characters in the movie.

Because the social network of a movie was a necessary condition for the inclusion of said movie into this study, only these movies were included.

## 3.2 Movie Success

The success of a movie was defined both according to the IMDb rating and according to the profit made by the movie in total worldwide. A successful movie rating-wise was defined as one having an IMDb score greater than or equal to 7.5. A successful movie profit-wise was defined as one having a gross worldwide revenue of at least two times its budget. If a movie met both of these conditions, it was labeled a success. If a movie met only one of these condition, it was labeled a partial success. Movies that met neither of these conditions were labeled as failures.

In order to find the relevant data, the list of movies in the social network database was extracted, and a web crawler built that searched through the IMDb pages of these movies for the rating and profit information. There were many movies for which rating, budget, and/or revenue information was not available. This decreased the size of the dataset to be analyzed from 15,538 movies in the original social network dataset to 6,472 movies in the final dataset.

## 3.3 Analysis

First, individual attributes were examined for their effect on movie success. Because the data of the attributes is not normally distributed, the Kruskal-Willis test was performed to evaluate statistical significance of the differences between movies that failed, partially succeeded, and succeeded. If statistical significance was established, a Dunn-Bonferroni post-hoc analysis was performed in order to determine pairwise significance of the effect of the attribute on the movie's success or lack thereof.

Next, an analysis of the interaction and subsequent effect of multiple social network attributes on movie success was performed. In order to evaluate the effect of more than one independent variable on the outcome of the movie success, a linear regression analysis was performed and p-values of the attributes examined to determine statistical significance of the effect of this attribute with the additional interaction between attributes.

Finally, using the same linear regression analysis, the effect of all of the social network attributes combined was evaluated to determine those attributes with the largest and most statistically significant effect on the success of the movie.

# 4 Results

## 4.1 Individual Attributes

When examining the effect of the individual attributes on movie success, all of these attributes had statistically significant effects on movie success except for the number of connected components. The Kruskal-Wallis test on the number of connected components produced a p-value of 0.57 whereas the p-values produced for all other attributes were less than 0.05. The fact that the number of connected components did not have a significant effect on movie success can also be seen by the fact that 95% of movies had only one connected component.

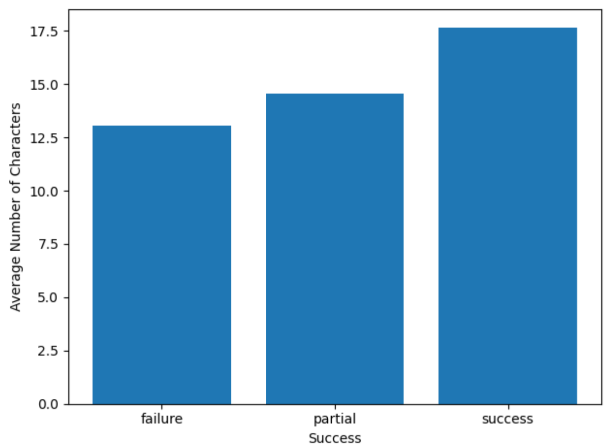


Figure 1: Average number of characters for different success levels of a movie. A failed movie had an average of 13.05 characters, a partially successful movie had an average of 14.57 characters, and a successful movie had an average of 17.64 characters. Statistical significance of the effect of number of characters on movie success was evaluated with the Kruskal-Wallis test with a p-value of $2.03 \times 10^{-34}$, and afterwards the Dunn-Bonferroni post-hoc was performed to verify that all p-values in the resulting matrix were less than 0.05. Graphs and subsequent averages and p-values for the other attributes can be found in the online python notebook.

Fig.1 shows the average numbers of characters for each of the different levels of movie success. The Kruskal-Wallis test and Dunn-Bonferroni post-hoc produced p-values of less than 0.05 also in the pairwise comparisons, meaning that the differences in average number of characters in each of the success levels were statistically significant. According to the average values, as can be seen in the graph, the number of characters in a movie has a significant effect on the success or failure of a movie, with the trend indicating that the more characters in a movie, the more likely it is to be successful.

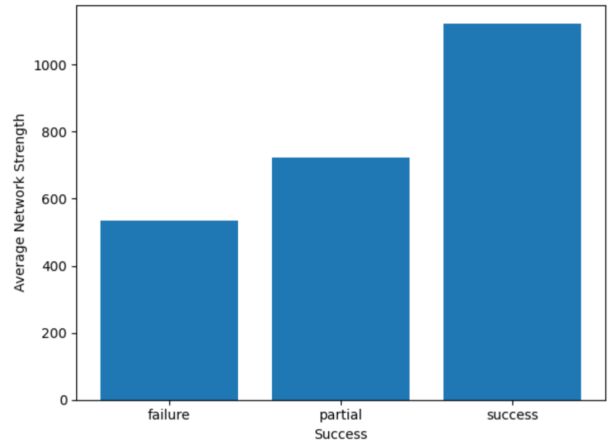


Figure 2: Average network strength for different success levels of a movie. A failed movie had an average of 535.77, a partially successful movie had an average of 721.66, and a successful movie had an average of 1120.98. Statistical significance of the effect of network strength on movie success was evaluated with the Kruskal-Wallis test with a p-value of $4.37 \times 10^{-39}$, and afterwards the Dunn-Bonferroni post-hoc was performed to verify that all p-values in the resulting matrix were less than 0.05. Graphs and subsequent averages and p-values for the other attributes can be found in the online python notebook.

Fig.2 shows the average network strength, the total sum of the edge weights of the network, for each of the different levels of movie success. The Kruskal-Wallis test and Dunn-Bonferroni post-hoc produced p-values of less than 0.05 also in the pairwise comparisons, meaning that the differences in average network strength in each of the success levels were statistically significant. According to the average values, as can be seen in the graph, the total number of character interactions in a movie has a significant effect on the success or failure of a movie, with the trend indicating that the more characters interact with each other and with other different characters in a movie, the more likely it is to be successful.

Fig.2 shows the average network strength, the total sum of the edge weights of the network, for each of the different levels of movie success. The Kruskal-Wallis test and Dunn-Bonferroni post-hoc both produced p-values of less than 0.05 for all comparisons, meaning that the differences in average network strength in each of the success levels were statistically significant. Thus, the total number of character interactions in a movie has a significant effect on the success or failure of a movie, with the trend indicating that the more characters interact with each other and with other different characters in a movie, the more likely it is to be successful.

The analysis of the effects of the rest of the individuals attributes on movie success was almost identical to the one performed on the number of characters with very similar plots produced. Both the Kruskal-Wallis and Dunn-Bonferroni post-hoc showed pairwise statistical significance of the differences between success levels for each attribute, with the trend indicating that the more of that attribute, the more likely the movie is to be successful. The rest of the graphs and data can found in the online notebook [1]

## 4.2 Combinations of Attributes

First, the effects of overall network connections were examined, where network connections are defined as those attributes that affect the entire network. Specifically, those attributes are number of characters, network strength, and network closeness. Fig. 3 shows the number of characters vs network strength with the points of the individual movies colored according to success. Fig. 4 shows a similar plot for network closeness vs network strength. By looking at these two plots, we can see that the more successful movies seemed to have a threshold of around 8000 for the network strength, with only failing or partially successful movies reaching higher values.

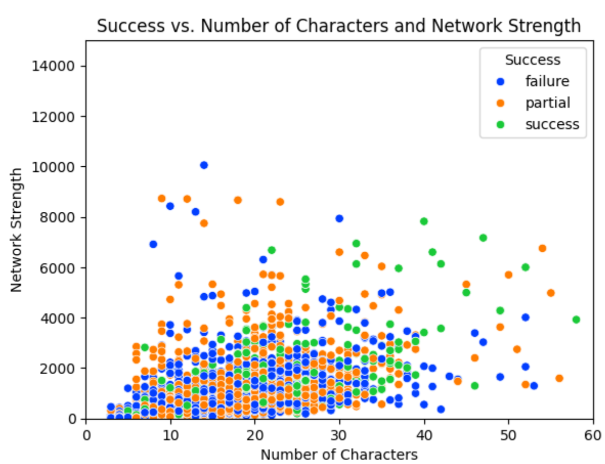


Figure 3: The number of characters in a movie plotted against the network strength with each movie colored according to its success level. The correlation coefficient calculated between number of characters and network strength was 0.6.

In order to examine the combined effects of these variables, a linear regression analysis was used and the p-values of each variable examined. When performing the analysis, the results showed that there was a strong co-linearity between the number of characters and network strength with a correlation coefficient of 0.6 and a strong co-linearity between the network closeness and the network strength with a correlation coefficient of 0.78. This indicates (quite logically) that as the number of characters increases, so does the total amount of interactions between characters, and as the number of interactions between unique pairs of characters increases, so does the total amount of interactions between characters. Thus, when examining the effect of all three of these variables on the success of the movie, only two of them were found to have a statistically significant effect: network strength and network closeness. However, because of the multicollinearity of these variables, the statistical significance of their combined effect is reduced.

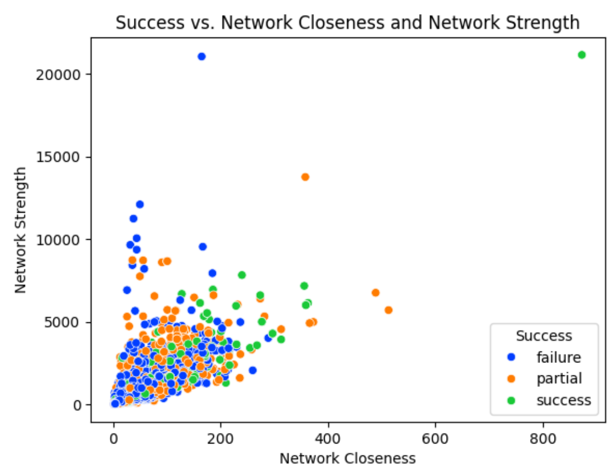


Figure 4: Network closeness of a movie plotted against the network strength with each movie colored according to its success level. The correlation coefficient calculated between network closeness and network strength was 0.78.

Second, the effects of the internal social dynamics of the network were examined, where these variables are the weight of the edge in the network with the maximum weight and the degree of the node in the network with the maximum degree. When performing the analysis, both the maximum number of interactions and the maximum node degree produced p-values of less than 0.05, indicating that there is a statistical significance to their effect on the suc-

[1] https://github.com/jordanco-bgu/social_network_movies/tree/main

cess of a movie when looking at the combination of their effects. Because the coefficient of the maximum number of interactions is less than the coefficient of the maximum node degree, we can conclude that the number of other characters the most social character interacts with has more of an influence on the success of a movie than the number of interactions between the two characters who interact the most in a movie.

If we look at Fig.5, it also seems as if the successful movies have a limit to the maximum number of interactions of around 1000, with only partially successful and failing movies having edge weights above this. Similarly, from the graph we can see the trend we saw in the individual analyses in that the more successful movies seem to have higher maximum node degrees. In addition, the number of connected components was added to this analysis, but there was no significant effect found of the number of connected components (social circles) in a movie's social network on the success of a movie.

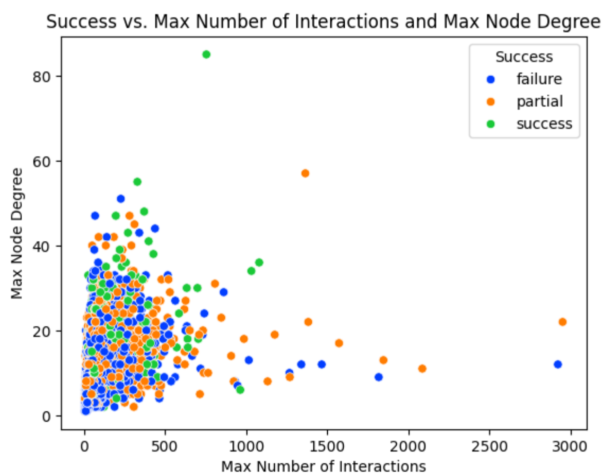


Figure 5: Maximum edge weight of a network plotted against the maximum node degree of the network with each movie colored according to its success level.

Lastly, we examined the effects of all attributes on the success levels of a movie. Because of the multicollinearity present between the network connection attributes, only network strength was chosen for this examination. In addition, the attribute of the number of connected components was removed because it had no significant effect on the success of the movie individually and when combined with the other social dynamics attributes. The attributes remaining that were analyzed together were: network strength, number of main characters, maximum edge weight, and maximum node degree.

When analyzing the p-values of these attributes, and thus the significance of their effect on the outcome, the network strength, maximum edge weight, and maximum node degree all had p-values of less than 0.05 while the number of main characters had a p-value of 0.755, indicating that it did not have a statistically significant effect on the outcome.

# 5 Discussion and Conclusions

Based on the individual attribute results and the combined attribute results, we can conclude that the three attributes of network strength, maximum edge weight, and maximum node degree have a significant impact on the success of a movie, and if movie producers wish to increase the likelihood of the success of their movies, they would do well to look at these three attributes. In general, the larger and more highly connected the social network and the more social the characters in the social network, the higher the likelihood of movie success.

In this study, multicollinearity was observed between network strength and number of characters and network strength and network closeness, indicating that they are influenced by each other. Further analysis is required in order to understand exactly which attributes are co-linear, why, and how we can reduce this effect while extracting the most important attributes contributing to the success of a movie. In addition, movies that were successful at the box office but not by rating and those who were successful by rating but not at the box office were categorized in the same way, as partially successful. A potential subject for future study is to understand if there is a difference between these two different kinds of "partially successful" and which attributes contribute the most to each. This could help to understand exactly which attributes contribute to which kinds of successes, if a trend like this exists at all.

In the course of this study, there were many different factors that were reviewed as possibly having effects on movie success, factors that were considered to be included in this study. However, in order to limit the scope of this initial exploratory study, it was decided to limit the attributes (and their effects) examined to those analyzed. The other factors considered, that could be used in further future research, are: effect of year or decade of the movie, number of celebrities involved in the making of the movie, country of origin of the movie and/or culture, and related movies such as sequels or those who belong to movie universes.

# References


[1] Ramesh Sharda and Dursun Delen. "Predicting box-office success of motion pictures with neural networks". In: *Expert Systems with Applications* 30.2 (2006), pp. 243–254.

[2] Barry Litman. "Predicting Success of Theatrical Movies: An Empirical Study". In: *Journal of Popular Culture* 16.4 (1983).

[3] Krushikanth Apala et al. "Prediction of movies box office performance using social media". In: (Aug. 2013), pp. 1209–1214.

[4] D Li and ZP Liu. "Predicting Box-Office Markets with Machine Learning Methods". In: *Entropy (Basel)* (2022).

[5] Dima Kagan, Thomas Chesney, and Michael Fire. "Using data science to understand the film industry's gender gap". In: *Palgrave Communications* (2020).